ADDRESS RESOLUTION PROTOCOL (ARP)

- Address Resolution Protocol (ARP) is a communication protocol used to find the MAC (Media Access Control) address of a device from its IP address.
- This protocol is used when a device wants to communicate with another device on a Local Area Network or Ethernet.
- ARP relates an IP address with the physical address. On a typical physical network such as LAN, each device on a link is identified by a physical address, usually printed on the network interface card (NIC).
- A physical address can be changed easily when NIC on a particular machine fails.

- The IP Address cannot be changed. ARP can find the physical address of the node when its internet address is known. ARP provides a dynamic mapping from an IP address to the corresponding hardware address.
- When one host wants to communicate with another host on the network, it needs to resolve the IP address of each host to the host's hardware address.

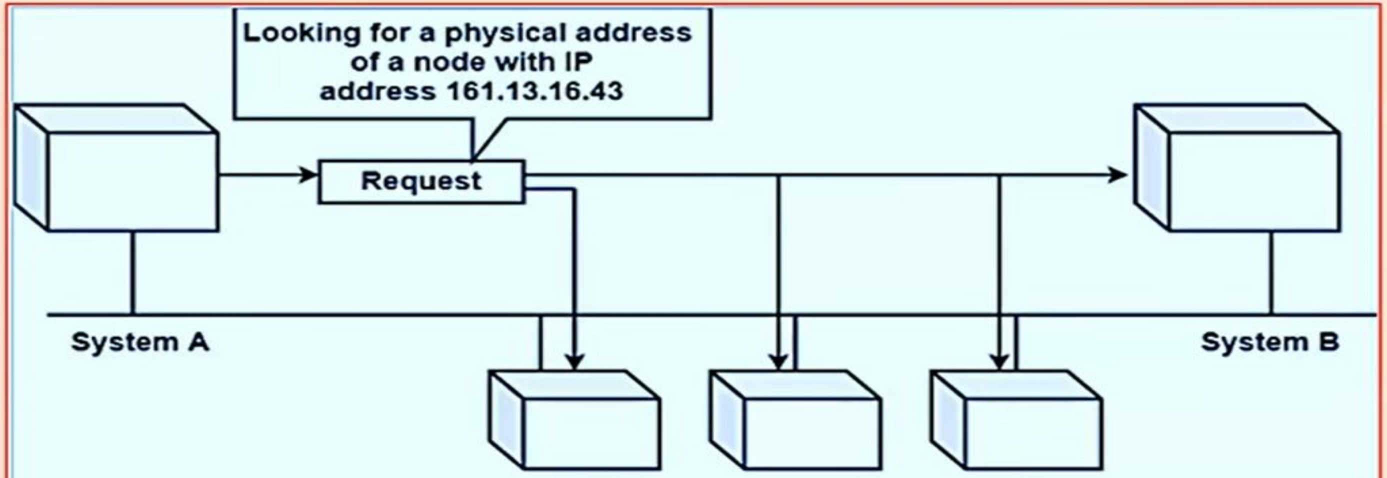
❖WORKING OF ARP

1. When a host tries to interact with another host, an ARP request is initiated. If the IP address is for the local network, the source host checks its ARP cache to find out the hardware address of the destination computer.

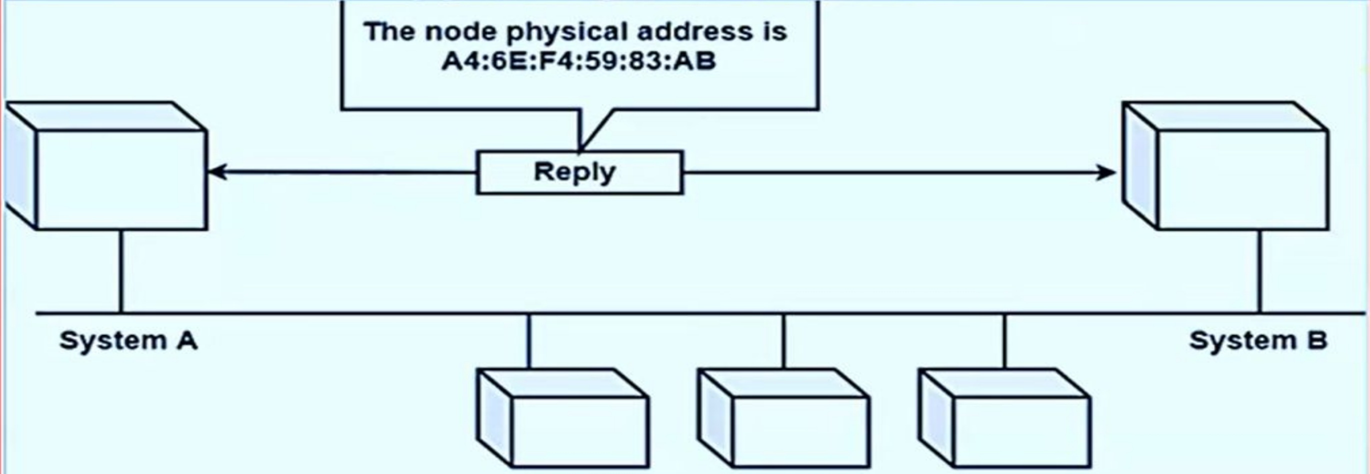
2. If the correspondence hardware address is not found, ARP broadcasts the request to all the local hosts.
3. All hosts receive the broadcast and check their own IP address. If no match is discovered, the request is ignored.
4. The destination host that finds the matching IP address sends an ARP reply to the source host along with its hardware address, thus establishing the communication.
5. The ARP cache is then updated with the hardware address of the destination host.

❖Important ARP terms

- **ARP Cache:** After resolving the MAC address, the ARP sends it to the cache stored in a table for future reference. The subsequent communications can use the MAC address from the table.
- **ARP Cache Timeout:** It is the time for which the MAC address in the ARP cache can reside.
- **ARP request:** Broadcasting a packet over the network to validate whether we came across the destination MAC address or not.
- **ARP response/reply:** The MAC address response that the source receives from the destination aids in further communication of the data.



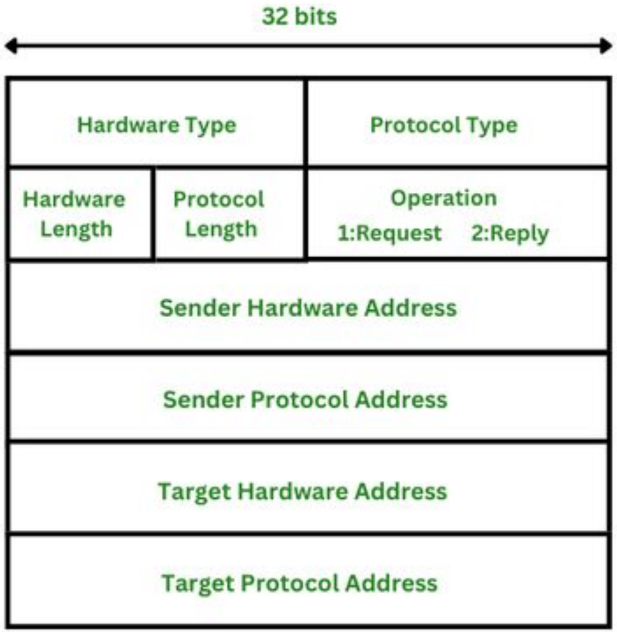
(a) ARP request is broadcast



(b) ARP reply is unicast

ARP Packet Format

The ARP packet format is used for ARP requests and replies and consists of multiple fields including hardware type, protocol type, hardware and protocol size, operation, sender and target hardware, and IP addresses. These fields work together to help devices on a network find and communicate with each other



Hardware type: This is 16 bits field defining the type of the network on which ARP is running. Ethernet is given type 1.

Protocol type: This is 16 bits field defining the protocol. The value of this field for the IPv4 protocol is 0800H.

Hardware length: This is an 8 bits field defining the length of the physical address in bytes. Ethernet is the value 6.

Protocol length: This is an 8 bits field defining the length of the logical address in bytes. For the IPv4 protocol, the value is 4.

Operation (request or reply): This is a 16 bits field defining the type of packet. Packet types are ARP request (1), and ARP reply (2).

Sender hardware address: This is a variable length field defining the physical address of the sender. For example, for Ethernet, this field is 6 bytes long.

Sender protocol address: This is also a variable length field defining the logical address of the sender For the IP protocol, this field is 4 bytes long.

Target hardware address: This is a variable length field defining the physical address of the target. For Ethernet, this field is 6 bytes long. For the ARP request messages, this field is all Os because the sender does not know the physical address of the target.

Target protocol address: This is also a variable length field defining the logical address of the target. For the IPv4 protocol, this field is 4 bytes long.